

Respiratory Exchange Ratio

RER staat voor Respiratory Exchange Ratio (ademschaalwisselingsratio). Het is de verhouding tussen de hoeveelheid uitgeademde kooldioxide CO₂ en de verbruikte zuurstof O₂:

$$\text{RER} = \text{VCO}_2/\text{VO}_2$$

Deze ratio geeft aan of het lichaam vooral vetten (RER=0,7) of koolhydraten (RER=1,0) verbrandt voor energie. In rust is de RER vaak rond de 0,8. Tijdens intensieve inspanning stijgt de RER richting of boven de 1,0, wat wijst op een groter beroep op anaerobe verbranding.

Omdat koolhydraten per liter zuurstof meer energie leveren (**5.05 kcal/liter O₂**) dan vetten (**4.69 kcal/liter O₂**), bepaalt de RER de exacte energetische waarde van zuurstof. De RER kan stijgen boven de 1.00 (tot wel 1.20). Dit wijst op anaerobe verbranding waarbij je lichaam meer CO₂ uitstoot dan het aan O₂ verbruikt om melkzuur te bufferen.

Technisch gezien spreken we hier over np-RER: er wordt bij de standaard RER-berekening abstractie gemaakt van proteïnen.

Belangrijkste redenen waarom proteïnen meestal buiten beschouwing worden gelaten:

- Beperkte bijdrage als brandstof: In een normale fysiologische toestand (en tijdens de meeste sportinspanningen) is de bijdrage van proteïnen aan de energievoorziening erg klein, meestal minder dan 5% tot 10%.
- Onvolledige verbranding: In tegenstelling tot koolhydraten en vetten worden proteïnen niet volledig verbrand tot CO₂ en H₂O. Het stikstofgedeelte kan niet door het lichaam worden geoxideerd en wordt uitgescheiden via de urine (als ureum). Dit maakt het onmogelijk om het exacte verbruik te bepalen met alleen een ademgasanalyse.
- Complexiteit: Omdat verschillende aminozuren op verschillende manieren worden gemetaboliseerd, is er geen eenduidige "vaste" waarde voor de verhouding CO₂/O₂ voor proteïnen (meestal wordt geschat rond de 0.81–0.82).

Als onderzoekers de verbranding van proteïnen wél mee willen rekenen, gebruiken ze indirecte calorimetrie in combinatie met een urineonderzoek: Ze meten de hoeveelheid stikstof in de urine over 24 uur. Hiermee berekenen ze hoeveel gram proteïne er is verbrand. Vervolgens trekken ze de O₂ en CO₂ die bij deze proteïneverbranding hoort af van de totale gemeten ademgassen. Wat overblijft is de niet-proteïne RER, die dan heel nauwkeurig de verhouding tussen vet- en koolhydraatverbranding weergeeft. Voor algemene toepassingen en commerciële fitness tests wordt dit bijna nooit gedaan omdat het verzamelen van urine omslachtig is en de foutmarge door het negeren van proteïnen klein genoeg is voor dagelijks gebruik.

Exact calorie burn: Indirect Calorimetry

The most precise way to measure calorie burn is indirect calorimetry, which measures the actual gas exchange (oxygen consumed and carbon dioxide produced) through a mask or metabolic chamber. It works because there is a direct chemical relationship between oxygen used and energy (calories) produced. It calculates the "Respiratory Exchange Ratio" (RER) to tell you exactly how many calories you are burning and whether they are coming from fat or carbohydrates.

From static HR values to VO2max: Uth-Sørensen-Overgaard-Pedersen estimation

The Uth-Sørensen-Overgaard-Pedersen estimation is a method to estimate maximal oxygen uptake (VO2max) using the ratio of maximal heart rate to resting heart rate for well-trained individuals.

$$VO_2\text{max}(\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}) = \text{PF} \times \frac{\text{HR}_{\text{max}}}{\text{HR}_{\text{rest}}} \quad \text{PF men} = 15.3 \text{ // PF women} = 14.5$$

The method shows a standard error of estimate (SEE) of around 4.5%, which is generally considered superior to other indirect, non-exercise tests. Using age predicted HR max increases error to 4.5%-7.8%

→ mijn horloge: 53 HRrest, 186 maxHR, VO2max calc = 53.7, VO2max volgens mijn horloge 63 (estimate based on HR & pace running outdoors)

Other, possibly better methods:

Accurate for Active Individuals: Cooper 12-Minute Run: The Cooper 12-Minute Run is widely considered one of the most valid field tests, showing a high correlation (0.90-0.92) with lab results. The Test: Run as far as possible in 12 minutes on a flat surface. It measures actual performance at near-maximal intensity, which more closely reflects true aerobic capacity than heart rate alone.

$$VO_2\text{max} = \frac{\text{Distance in meters} - 504.9}{44.73}$$

For cycling/cross training: Highly Accurate for Cyclists: 5-Minute Power Test. If you use a power meter, a 5-minute all-out effort can estimate VO2max with a remarkably low 3% error rate 95% of the time.

$$VO_2\text{max} (\text{ml/min/kg}) = 16.61 + 8.87 \times \text{Power Output (W/kg)}$$

As you climb higher, the air becomes less dense. While this makes breathing harder (reducing your VO2max), it actually lowers your energy cost of moving forward because there is less air resistance (drag). h staat voor hoogte en k is een constante (0.0001–0.00015)

$$VO_2^{\text{max}}(h) = VO_2^{\text{max}}(0) \cdot e^{-kh}$$

Obtain BMR: Mifflin-St Jeor eq, katch-McArdle eq and Cunningham eq

All three formulas give BMR in kcal per day. The **Mifflin-St Jeor eq.** is currently the industry standard for people who do not know their body composition.

$$BMR = (10 \times \text{weight in kg}) + (6.25 \times \text{height in cm}) - (5 \times \text{age in years}) + s$$

s = + 5 for men // s = -161 for women // s = -78 for trans

But more muscle means more calories so somewhere fat % should make sense to include? The **Katch-McArdle eq.** is the most accurate formula when Lean Body Mass (LBM) is known. It ignores gender, height, and age because muscle mass is the primary driver of metabolic rate.

$$LBM: LBM = \text{Total Weight (kg)} \times (1 - \text{Body Fat \%}).$$

$$BMR: 370 + (21.6 \times LBM).$$

Lean Body Mass (LBM): The total weight of the body minus the weight of fat mass (Weight - Fat Mass). It is used for both men and women as it is based on active tissue rather than gender-specific averages.

In elite sports science, the **Cunningham Eq.** is often considered slightly more accurate than Katch-McArdle for highly active individuals and athletes. It uses a higher constant and multiplier because it was derived from a population of athletes with higher resting metabolic demands.

$$\text{Formula: } BMR = 500 + (22 \times \text{Lean Body Mass in kg})$$

or 492 in revised Cunningham eq.

What to use: very fit: Cunningham, DEXA scan or skinfold caliper test // relatively fit: Katch-McArdle // otherwise use Mifflin-St Jeor

From BMR to MET reference

The "Standard MET" (3.5 mlO₂/min/kg) is an average based on a 70kg, 40-year-old male.

Calculate personalized 1 MET:

- Step 1: Calculate Daily BMR (Mifflin-St Jeor or Katch-McArdle)
- Step 2: Divide by 1440 to get BMR/minute
- Step 3: Divide by 4.82 (assumption: 1 liter of O₂ = 4.82 kcal - resting RER of 0.82), then multiply by 1000 to obtain resting VO₂ in ml/min
- Step 4: Divide by weight in kg to get personalized 1 MET value

From heart rate to calories: Keytel formula

It uses a regression equation that factors in your heart rate alongside your biometrics. The Logic: Your heart pumps blood to deliver oxygen to your muscles. As exercise intensity increases, your heart rate (HR) rises linearly (assumption) with your oxygen consumption (VO₂).

This formula does not take fitness into account:

for men: $EE(kJ/min) = -55.0969 + (0.6309 \times HR) + (0.1988 \times W) + (0.2017 \times A)$

for women: $EE(kJ/min) = -20.4022 + (0.4472 \times HR) - (0.1263 \times W) + (0.074 \times A)$

HR = heart rate, W = gewicht in kg, A = leeftijd in jaar (conversie kJ to kcal is delen door 4.184)

This variant takes fitness into account:

$$EE(kJ/min) = -95.7735 + (0.634 \times HR) + (0.404 \times VO_{2max})$$

men: $+(0.394 \times W) + (0.271 \times A)$

$$EE(kJ/min) = -59.3954 + (0.45 \times HR) + (0.380 \times VO_{2max})$$

women: $+(0.103 \times W) + (0.274 \times A)$

shortcoming general mass (calc):

Keytel uses total weight, this overestimates the "idling cost" for individuals with higher body fat and underestimates it for muscular athletes. Calculate the "Composition Gap":

1. Find your Muscular BMR (Katch-McArdle or cunningham, which uses lean mass).
2. Find the Average BMR for your weight (Mifflin-St Jeor, which Keytel effectively assumes).
3. The composition Gap = (Muscular BMR - Average BMR) / 1440.
4. Apply to Keytel:

$$\text{Final } EE = \text{Keytel Result} + \frac{(\text{Katch BMR} - \text{Mifflin BMR})}{1440 \times 4.184}$$

4.184 is needed as BMR is in kcal
or

$$\text{Final } EE = (EE_{raw} - RMR_{proxy}) + RMR_{true}$$

with $RMR_{proxy} = a \times W + b \times A + C$ in keytel and
 $RMR_{true}(kJ/min) = \frac{370 + (21.6 \times \text{Lean Mass})}{1440} \times 4.184$ 4.184 is needed as BMR is in kcal

shortcoming temperature (calc):

Ambient temperature significantly skews the heart rate/oxygen relationship. Heart rate increases by roughly 1 beat per minute (bpm) for every 1°C increase in ambient temperature above 24°C as HR rises to pump blood to the skin for cooling, not for muscle work. Use an Arrhenius-inspired HR correction. For every 1°C above 24°C, subtract 1.5% of HR.

$$HR_{adj} = HR_{raw} \times (1 - 0.015 \times (\text{Temp}_{ext} - 24))$$

glycogen depletion (calc):

Keytel implicitly assumes 20.9 kJ/LO₂ or 5.05 kcal/LO₂ with perfect carb burn. But even with BMR, if you go long, you deplete carbs and turn to fats → use a sliding scale from Lowering the efficiency factor reflects that your "metabolic motor" is working harder to get the same kJ of energy from fat.

$$RER(t) = RER_{start} \cdot (1 - \delta \cdot \frac{t}{t_{glycogen}})$$

$$M_{fuel} = \frac{(1.1 \cdot RER(t) + 3.9)}{5.05}$$

met t_{glyco} hoelang je glyco voorraad hebt (=vanaf daarna 100% vet) en met sigma hoe snel je door de voorraad heen zit (0.1 goed gefueled - 0.3 praktisch geen carbs meer). 1.1, 3.9 en 5.05 zijn chemische constanten voor zuurstofverbranding.

$$\text{Corrected } EE = [(EE_{raw} - RMR_{proxy}) + RMR_{true}] \times M_{fuel}$$

static body mass (calc):

Update mass dynamically: mass = mass_at_start - (used_energy - eaten_energy) - (used_fluid - taken_fluid)

shortcoming HR drift (calc or factor):

Cardiac Drift means HR–VO₂ Decoupling. Keytel assumes a stable HR–VO₂ relationship, which breaks down in long events. During ultras/long events: HR gradually rises without increased output, which is caused by dehydration, heat stress, glycogen depletion, fatigue (= Cardiac Drift). Keytel will overestimate kcal late in races, especially after ~2–3 hours.

→ difficult to account for

Factor: In events longer than 2 hours, HR drift typically accounts for a 5-15% inflation in HR that does not correspond to VO₂. Multiply the final Keytel EE by a decay factor: $D = 1 - 0.02 \cdot \text{hours_passed}$ → does not work for very long races cause then too low

Calc: use reference, average HR and effort (adjust pace eg) from start to actual to see drift, patterns in data. In the first 20–30 minutes (after warm-up, before drift starts), find a steady section and calculate speed/HR = R_{now} . then later recalculate $R = \text{pace/speed}$. Important to use intensity adjusted pace by elevation for example. Drift percent is then:

$$D = (1 - \frac{R_{now}}{R_{base}}) \times 100 \quad \text{such that} \quad HR_{adj} = HR_{now} \times (1 - D)$$

But in very long races: this method gets tricky because the drop in "meters per beat" is caused by two different things:

- Cardiac Drift: Heart beats faster for the same oxygen (Overestimates calories).
- Economy Decay: Muscles need more oxygen for the same speed (Underestimates calories).

In the first 4 hours: Assume the drop is 100% Cardiac Drift. Correct the HR downward. After 6–8 hours: Assume the drop is 50% Drift / 50% Muscle Fatigue. Only correct the HR by half of the calculated D. This accounts for the fact that you actually are burning more fuel because your running form has "broken."

shortcoming night HRV or real time (factor):

Integrating night HRV manually is essentially adding a "Fatigue Tax." It acknowledges that a beat-per-minute is more "expensive" when the heart is stressed (low HRV) than when it is relaxed.

Keytel captures lower HRV slightly. But if your HRV is in the "Red" (very low), add an extra 3–5% to the Keytel result to account for the biomechanical inefficiency (muscle friction) that even heart rate can't fully see.

In Summary:

1. **Correct the Input:** $HR_{adj} = (HR_{raw} - \text{Temp Correction}) \times (1 - D)$.
2. **Run Keytel:** Calculate EE_{raw} using HR_{adj} , VO_{2max} , W , and A .
3. **Personalize Biology:** $EE_{engine} = EE_{raw} + G$ (The BMR Gap).
4. **Personalize Chemistry:** $Final\ EE = EE_{engine} \times M_{fuel}$.

This gives Energy per minute

VO2 use during running: Léger & Mercier equation & Minetti

In their 1984 research comparing track vs. treadmill running, **Léger and Mercier** found that for horizontal track running at speeds between 8 and 25 km/h, the relationship between oxygen uptake and speed is almost perfectly linear. They established that 1 km/h of running speed requires exactly 1 MET of oxygen. Léger & Mercier's formula essentially assumes that for every 1 km/h of speed, you consume exactly one resting-unit of oxygen.

VO2 (mlO2/kg/min) = personalised MET * speed (--> general formula = 3.5*speed)

But only works on 'flat' terrain and 'short' distances. → Minetti!

Minetti's 5th-order polynomial was specifically designed to be accurate on extreme inclines where HR-based formulas (like Keytel) lose their linear relationship with oxygen cost.

cost of transport: $C_r = 155.4i^5 - 30.4i^4 - 43.3i^3 + 46.3i^2 + 19.1i + 3.6$ unit= J/kgm,
i=gradient in decimal. Maar opnieuw problemen:

the BMR 3.6 is generic (calc):

Replace 3.6 in the Minetti equation with Cpers. This anchors the formula to Katch-McArdle BMR.

$$C_{pers} = \frac{\text{Personal 1 MET} \times 20.9}{\text{Speed (m/min)}}$$

calibrated on asphalt (calc):

Form and terrain change your energy demand in ways no current watch can perfectly track. Vertical oscillation (bouncing) and ground contact time explain about 4–12% of individual variance in energy cost. Running on mud, sand, or technical trails requires more energy for stabilization. Multiply the final result by a Terrain Coefficient (N).

- Asphalt/Treadmill, high cadence/low bounce = no adjustment: N = 1
- Hard-packed Trail: N = 1.05
- Technical/Rooty Trail: N = 1.12
- Soft Sand/Deep Mud: N = 1.2-1.5

"Bouncy" Stride/Technical Terrain: Add 5-10% to the Minetti mechanical cost. Technical trails (mud, roots) increase energy demand by up to 20% because your stabilizing muscles are working overtime.

economy decay: you start to run bad (calc):

Running Economy (RE) is not a constant; it deteriorates over time due to muscle damage and tendon fatigue; your muscles lose elasticity. In ultra-marathons, the energy cost of running can

increase by 12% to 18% over the course of the race. In Ultras, apply a Time-Based Decay (ED). This accounts for the fact that every meter of climbing "damages" your efficiency for the rest of the race.

$$ED = 1 + \min(0.25, [\text{Time}_{tax} + \text{Vertical}_{tax} + \text{Impact}_{tax}])$$

Time tax: 0-12u +0.005/h // 12-24u +0.0025/h // >24u cap at 0.1

Vertical tax: +0.01 for every 300D+

Impact tax: +0.02 for every 300D-

Total Cap: 0.25

Now the formula becomes $C_{final} = C_r \times \eta(\text{Terrain}) \times ED(\text{Decay})$ with also personal BMR

then $VO_{2_total}(ml/kg/min) = \left(\frac{C_{final}}{20.9} \times \text{Speed (m/min)} \right)$ 20.9 shows how much oxygen is needed to do the required work, and it is not a constant! As you use more fat, you burn more oxygen so use variable factors.

shortcoming glycogen depletie (calc):

Glycogen (carbohydrates) is a more efficient fuel than fat. It produces about 12% more energy per liter of oxygen. When you "bonk" or run out of glycogen, your body switches to burning almost 100% fat. Even if your heart rate stays the same, your Oxygen Cost increases just to produce the same power output.

Intensity Level	Estimated RER	Personalised Joules/ml	Use Case
All fat		4.69	
Very Low (<60% HRmax)	0.75	20.0 = 4.780	Ultra-marathon walking/easy hiking
Moderate (60-80% HRmax)	0.85	20.3 = 4.852	Ironman bike/run, steady trail run
High (80-90% HRmax)	0.95	20.9 = 4.995	Half-marathon, hard trail efforts
Maximal (>90% HRmax)	1.00	21.1 = 5.043	5k/10k racing, sprint finishes
All carbs	1.00	5.05	

Use a sliding scale kcal factor. Start at 5.05 and gradually reduce it toward 4.8 → HR can indicate carbs but they might be empty and you use more fat. RER can be always calculated as:

$$RER = 0.7 + 0.3 \cdot \frac{VO_2}{VO_2^{max}}$$

An estimation on this formula is the Weir eq which uses RER at the start (basically 0.95-1), t_glyco hoelang je glyco voorraad hebt, en met sigma (=rate of depletion) hoe snel je door de voorraad

heen zit (0.1 = iron man goede fueling - 0.3 brute ultra)

$$RER(t) = RER \cdot \left(1 - \delta \cdot \frac{t}{t_{glycogen}} \right)$$

Dan komen we tot volledige energieberekening:

$$\text{kcal/min} = \frac{VO_{2_total} \times \text{Weight}}{1000} \times (1.1 \cdot RER(t) + 3.9)$$

In Summary:

$$C_r = (155.4i^5 - 30.4i^4 - 43.3i^3 + 46.3i^2 + 19.1i) + C_{pers}$$

N = terrain index, ED = fatigue index

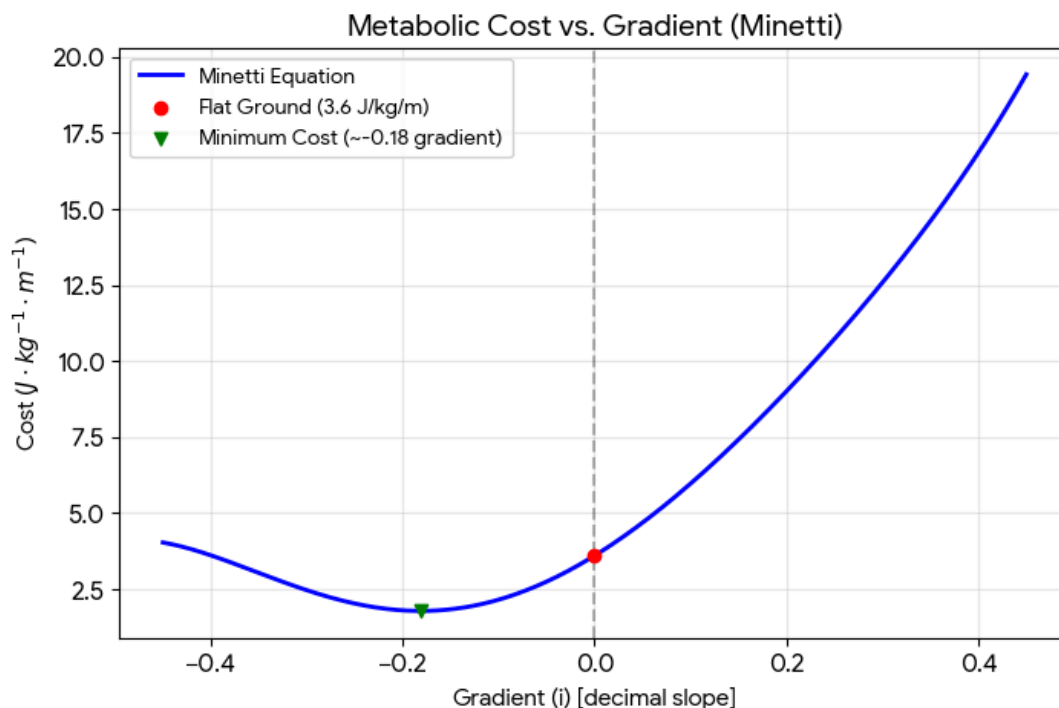
$$VO_{2_total} = \left(\frac{C_r \cdot \eta \cdot ED}{F(t)} \times \text{Speed (m/min)} \right)$$

Work to oxygen:

with F = fuel factor

$$\text{kcal/min} = \frac{VO_{2_total} \times \text{Weight}}{1000} \times (1.1 \cdot RER(t) + 3.9)$$

And finally



Conclusions (for now)

Energie (kcal) = time * VO2 (liter/min) * kcal/liter O2 (1 liter O₂ ≈ 4.8–5.05 kcal, afhankelijk RER)

VO2 (liter/min) = VO2_BMR + VO2_ACTIVE // RER = 0.7 + 0.3 * VO2/VO2max

Comparison

	Keytel Equation	Minetti Equation
Primary Data	Heart Rate (HR), Age, Weight.	Slope (GPS/Grade), Speed, Weight.
Metric Type	Internal Load: Measures how hard your "engine" is working.	External Load: Measures the mechanical cost of the task.
Philosophy	If the heart beats faster, more energy is being used.	Moving mass X up slope Y at speed Z costs energy Q

Keytel (HR-based): Does not calculate VO2 as an intermediate step; it maps biometrics directly to calories. It captures biological "hidden" costs: heat, dehydration, and mental fatigue.

Minetti (VO2-based): Calculates the required VO2 to overcome gravity. It is a mathematical ideal. It tells you what the energy cost should be if you were a perfectly efficient machine.

Keytel wins on fatigue,

Minetti wins on hills

Keytel: Fails if HR is skewed by caffeine, stress, or heat (where HR rises without a proportional increase in oxygen/calorie use). It is also inaccurate at very low (<90 bpm) or maximal intensities.

Minetti: Fails to account for very technical terrain. It assumes a smooth surface. Running on mud, loose rock, or sand can increase energy cost by 20–50%, which Minetti's "smooth" polynomial ignores.

If Keytel Calories >> Minetti Calories: You are likely overheating, dehydrated, or highly fatigued. Trust Keytel for total burn, but know your efficiency has dropped.

If Minetti Calories >> Keytel Calories: This often happens on very steep technical climbs where you move slowly. Trust Minetti here, as HR often fails to capture the high muscular (anaerobic) cost of steep hiking.

The final conclusion: Use Keytel for the "big picture" energy burn, but override it with Minetti calculations for steep, technical climbing sections where heart rate doesn't tell the full mechanical story.

$$\text{Efficiency} = \frac{\text{Energy Minetti (The Work)}}{\text{Energy Keytel (The Cost)}}$$

If the Gap closes: You are becoming a more efficient runner. If the Gap widens: You are losing "Economy." This is a signal to check your form, your fueling, or your fatigue levels.

Method summarised:

The Personal "Anchor" (Pre-Race Calibration)

Calculate your biological constants based on Lean Body Mass (LBM):

- determine BMR via katch-McArdle eq and Cunningham eq
 - get personal 1 MET
 - get composition Gap: muscular BMR - normal BMR (Mifflin)
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STEP 2: The Two Calculation Engines

Advanced Keytel (Internal Load): captures heart stress, heat, dehydration, and systemic fatigue.

1. **Correct the Input:** $HR_{adj} = (HR_{raw} - \text{Temp Correction}) \times (1 - D)$.
2. **Run Keytel:** Calculate EE_{raw} using HR_{adj} , VO_{2max} , W , and A .
3. **Personalize Biology:** $EE_{engine} = EE_{raw} + G$ (The BMR Gap).
4. **Personalize Chemistry:** $Final\ EE = EE_{engine} \times M_{fuel}$.

Advanced Minetti (External Load): captures gravity, terrain technicality, and mechanical muscle damage.

1. **Calculate C_r :** Polynomial cost using i (slope) + your C_{pers} .
$$C_r = (155.4i^5 - 30.4i^4 - 43.3i^3 + 46.3i^2 + 19.1i) + C_{pers}$$
 2. **Apply Modifiers:** $C_{total} = C_r \times \eta(\text{Terrain}) \times ED(\text{Economy Decay})$
 - $ED = 1 + \min(0.25, [T_{tax} + V_{tax} + I_{tax}])$
 3. **Convert to Oxygen:** $VO_{2_total} = (C_{total}/F(t)) \times \text{Speed (m/min)}$
 - $F(t) = 21.1 \rightarrow 19.6$ based on $RER(t)$.
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STEP 3: The Fuel Efficiency Filter

Adjusts for the shift from Glycogen to Fat using the Weir Equation.

1. **Dynamic RER:** $RER(t) = 0.95 \cdot (1 - \delta \cdot \frac{t}{t_{glycogen}})$
 2. **Kcal Multiplier:** $M_{fuel} = (1.1 \cdot RER(t) + 3.9)/5.05$
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STEP 4: The Final Integration

To find the Total Energy for any segment of the race:

Scenario	Rule	Logic
Flat & Steady	Use Keytel $\times M_{fuel}$	HR is the most stable proxy here.
Steep/Technical	Use Minetti $\times M_{fuel}$	Gravity/Terrain dominate over HR.
Late Race (12h+)	Use Minetti $\times M_{fuel}$	HR suppression/drift makes HR data "noisy."
High Heat	Compare Both	If Keytel $\gg$ Minetti , you are overheating.

$$\text{Efficiency} = \frac{\text{Energy Minetti (The Work)}}{\text{Energy Keytel (The Cost)}}$$

If the Gap closes: You are becoming a more efficient runner. If the Gap widens: You are losing "Economy." This is a signal to check your form, your fueling, or your fatigue levels.

Required parameters:

Parameter	Type	Source/tool
Weight & height	static	scale / measurement
Body fat %	static	DEXA scan or calipers
VO2max	static	Lab test, 12min run test or other mentioned tests
Glycogen benchmarks (time to depletion, depletion rate)	static	Historical fueling (I hit wall after ... time) or standard values
HR	dynamic	chest strap (best) or watch
Distance, speed & slope	dynamic	GPS / barometric altimeter
External temperature	dynamic	Watch sensor or weather app

The total energy calculation after a race:

this takes after burn (EPOC) into account and integrates the method over time

$$\text{Final kcal} = \left(\frac{\text{Selected Method (kJ/min)}}{4.184} \right) \times \text{Duration} \times 1.05(\text{EPOC})$$

What official watches: This is the "Pro-Level" benchmark for field-based estimation. However, to answer your question with total transparency: No, this is not the absolute ceiling of what is possible, but it is the absolute limit of what you can do without a laboratory.

1. What "Top Brands" (Garmin, Polar, Apple) Use

Top brands do not use Minetti or Keytel directly. They use Proprietary Machine Learning Models (most famously Firstbeat Analytics, owned by Garmin).

- **Heart Rate Variability (HRV):** High-end watches don't just look at beats per minute (Keytel); they look at the *millisecond timing* between beats. This allows them to estimate **EPOC** (Excess Post-exercise Oxygen Consumption) and real-time VO_2 much more accurately than a linear formula.
- **The "Black Box":** These brands have databases of tens of thousands of lab tests. They use neural networks to recognize patterns. If your HR is high but your pace is slow, the watch "guesses" you are on a hill or fatigued based on historical data patterns.
- **Weakness:** These are still estimates. They often struggle with "Trail" because GPS distance on switchbacks is inaccurate, which throws off their internal calculations.

2. What is "Better" than our Hybrid Approach?

If you wanted to move from 90-95% accuracy to 99%, you would need to eliminate the last few "assumptions" by adding hardware:

- **Running Power Meters (Stryd):** Instead of using Minetti (which assumes a smooth slope), a power meter uses accelerometers to measure the actual mechanical force you exert against the ground 50 times per second. It accounts for wind resistance and "mushy" ground (sand/mud) which Minetti cannot see.
- **Core Temperature Sensors:** These account for "Heat Strain." If your HR is high because your internal temp is 39°C, a sensor tells the formula to stop counting those extra beats as "work calories" and start counting them as "cooling calories" (which have a different metabolic cost).
- **Portable Metabolic Carts (POGO):** This is the actual "Best of the Best." You wear a mask during the race. It measures every molecule of CO_2 you exhale. This is the only way to get a 100% "No Assumptions" calorie count.

3. Why the "Keytel-Minetti Hybrid" is still your best bet

For a person without a mask or a lab, the approach we built is superior to a standard Garmin for one reason: **Transparency**. Garmin might give you a number, but you don't know *why*. Your Hybrid Method allows you to see the "gap" between internal effort (HR) and external work (Slope). That gap is the most valuable data point in sports science—it tells you your Efficiency.

The Final Verdict

This is the Gold Standard for Manual Estimation. To go any further, you would have to stop using *formulas* and start using *sensors* (Power meters and Core temp). For 99% of elite athletes, the Keytel + Personalized BMR + Minetti logic provides more than enough precision to manage fueling for an Ironman or an Ultra.